

Computation of a_2

The code for a_2 effectively computes a geometric series. In fact one finds

$$f(n) = \sum_{k=3}^{n+2} (1/2)^k - (\text{negligible terms}),$$

so as $n \rightarrow \infty$, $f(n) \rightarrow (1/2)^3 / (1 - 1/2) = 1/8 \div (1/2) = 1/4$ (using the standard infinite geometric-series formula ¹). Hence

$$2f(n) + \frac{1}{2} + 0.25^n \rightarrow 2 \cdot \frac{1}{4} + \frac{1}{2} + 0 = 1.$$

The tiny corrections $2 \cdot 10^{-24}k$ in the code are utterly negligible, so in the limit a_2 is just above 1 . Numerically $a_2 \approx 1.00$ to many decimals.

Optimization problem

We must minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$. This is a simple linear program. It is well known that a linear objective attains its optimum at a vertex of the feasible region ². Here the feasible set (the line segment) has only two extreme points:

- $x_1 = 1, x_2 = 0$ (using only therapy 1), and
- $x_1 = 0, x_2 = 1/a_2$ (using only therapy 2).

We compare the total time at these vertices: at $(1, 0)$ the time is $1 + 0 = 1$, at $(0, 1/a_2)$ the time is $0 + 1/a_2 = 1/a_2$. The smaller of 1 and $1/a_2$ gives the minimizer.

Optimal solution and values

Since a_2 (by the code) is just above 1 , we have $1/a_2 < 1$. Therefore the minimum is attained at $x_1 = 0, x_2 = 1/a_2$. Numerically $1/a_2 \approx 1.00$ (to two decimal places). Thus an optimal solution (to within 0.005 in each component) is

$$x_1 \approx 0.00, \quad x_2 \approx 1.00.$$

In other words, one should use essentially the second (slightly faster) therapy only. This choice indeed minimizes the total time.

Answer: $x_1 \approx 0.00, x_2 \approx 1.00$.

Explanation: As above, the coded value of a_2 approaches 1 (from above) by geometric-series summation ¹, so the higher dose rate is $a_2 > 1$; hence the fastest delivery is to spend all time on therapy 2. By linear-program theory the optimum lies at one endpoint ². (We checked that $1/a_2 < 1$, so $(0, 1/a_2)$ beats $(1, 0)$.)

1 Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series

2 A Lower Bound On Determinants From Linear Programming

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